

DR. BABASAHEB AMBEDKAR TECHNOLOGICAL UNIVERSITY, LONERE

Bachelor of Technology (Civil and Infrastructure Engineering) SEMESTER - 5 Winter 2025 (Supply.)

Course :Bachelor of Technology (Civil and Infrastructure Engineering) Branch : Engineering and Technology

Semester : SEMESTER - 5

Subject Code & Name: BTCIC502 - SOIL MECHANICS & FOUNDATION ENGINEERING

Time : 3 Hours]

[Total Marks : 60

Instructions to the Students:

1. Each question carries 12 marks.
 2. Question No. 1 will be compulsory and include objective-type questions.
 3. Candidates are required to attempt any four questions from Question No. 2 to Question No. 6
 4. Use of non-programmable scientific calculators is allowed.
 5. Assume suitable data wherever necessary and mention it clearly.
-

Q1. Objective type questions. (Compulsory Question) 12

- 1 According to Terzaghi's effective stress principle, the effective stress in soil is given by:
A) Total stress + pore water pressure
B) Total stress – pore water pressure
C) Pore water pressure – total stress
D) Neutral stress – total stress
- 2 In a fully saturated soil mass, the number of phases present is:
A) One
B) Two (solid and water)
C) Three (solid, water, air)
D) Four
- 3 An undisturbed soil sample is essential for determining:
A) Grain size distribution
B) Atterberg limits
C) Shear strength and compressibility
D) Specific gravity
- 4 The unit weight of soil solids depends mainly on:
A) Void ratio
B) Water content
C) Specific gravity of solids
D) Degree of saturation
- 5 Field density of soil is most accurately determined using:
A) Hydrometer method
B) Core cutter method
C) Pycnometer method
D) Sieve analysis
- 6 Plasticity index of soil is numerically equal to:
A) $LL + PL$
B) $LL - PL$
C) $PL - SL$
D) $LL - SL$

- 7 Coulomb's failure criterion for soil is expressed as:
 - A) $\tau = \sigma$
 - B) $\tau = c + \sigma \tan \phi$
 - C) $\tau = \sigma \tan \phi$
 - D) $\tau = c - \sigma \tan \phi$
- 8 Which shear test is most suitable for determining undrained shear strength of soft clay in the field?
 - A) Direct shear test
 - B) Triaxial CD test
 - C) Vane shear test
 - D) Consolidated drained test
- 9 Safe bearing pressure of soil considers:
 - A) Ultimate bearing capacity only
 - B) Factor of safety only
 - C) Settlement criteria only
 - D) Both shear failure and settlement
- 10 A raft foundation is generally preferred when:
 - A) Soil bearing capacity is very high
 - B) Column loads are small
 - C) Soil is weak and differential settlement is likely
 - D) Groundwater table is very deep
- 11 Factor of safety of a slope is defined as the ratio of:
 - A) Driving force to resisting force
 - B) Resisting force to driving force
 - C) Shear stress to normal stress
 - D) Cohesion to friction
- 12 Stability number is primarily used for:
 - A) Bearing capacity analysis
 - B) Earth pressure calculation
 - C) Slope stability of cohesive soils
 - D) Settlement analysis

Q2. Solve the following.

- A) The water table in a deposit of sand 8 m thick, is at a depth of 3 m below the surface. Above the water table, the sand is saturated with capillary water. The bulk density of sand is 19.62 kN/m^3 . Calculate the effective pressure at 1 m, 3 m and 8 m below the surface. Hence plot the variation of total pressure, neutral pressure and effective pressure over the depth of 8 m.
- B) Draw the neat sketch of soil sampler showing its components. Explain the below 6 terms associated with it:
 - 1) Area Ratio, 2) Inside Clearance & 3) Outside clearance

Q3. Solve the following.

- A) What are Atterberg Limits? Explain their significance in soil classification. 6
- B) Explain the Following terms: 6
 - 1) Voids Ratio 2) Porosity 3) Degree of saturation
 - 4) Water Content 5) Specific Gravity 6) Air Content

Q4. Solve Any Two of the following.

- A) State Darcy's Law and discuss the factors affecting permeability. 6

- B) Describe Falling Head Permeability Test with proper diagram. 6
- C) Calculate the coefficient of permeability of a soil sample, 6 cm in height and 50 cm² in cross-sectional area, if a quantity of water equals to 430 ml passed down in 10 minutes, under an effective constant head of 40 cm. On oven-drying, the test specimen has mass of 498 g. Taking the specific gravity of soil solids as 2.65, calculate the seepage velocity of water during the test. 6
- Q5. Solve Any Two of the following.
- A) Explain the concept of Safe Bearing Pressure and discuss the factors affecting it. 6
- B) Explain the sheet pile walls in detail using neat sketches. 6
- C) Draw a neat, labelled sketch of typical section of well foundation. Also, draw 6 neat sketches of common types of shapes of well foundations.
- Q6. Solve Any Two of the following.
- A) What are the different types of slopes. Explain in detail with neat sketch and 6 example?
- B) Enlist and explain in detail the different types of slope failures? 6
- C) A slope is to be constructed at an inclination of 30° with the horizontal. Determine 6 the safe height of the slope at factor of safety of 1.5. The soil has the following properties: $c = 15 \text{ kN/m}^2$, $\phi = 22.5^\circ$ and $\gamma = 19 \text{ kN/m}^3$

*** End ***